

proprietary interfaces. There may be one EM per VNF or one EM can manage multiple VNFs.

3. *VNF manager (VNFM)*: This manages one or more VNFs, i.e., it does the life cycle management of VNF instances, which includes setting up/maintaining and tearing down VNFs. Additionally, VNFM does the FCAPS for the virtual part of the VNF.

The difference between the EM and VNFM is that the former manages functional components, while the VNFM manages the virtual components.

4. *NFVI (network function virtualisation infrastructure)*:

NFVI is the environment in which VNFs run. This includes physical resources, virtual resources and the virtualisation layer, described below.

- *Compute, memory and networking resources*: This is the physical part in NFVI. Virtual resources are instantiated on these physical resources.
 - *Virtual compute, virtual memory and virtual networking resources*: This is the virtual part in NFVI. The physical resources are abstracted into virtual resources that are ultimately used by VNFs.
 - *Virtualisation layer*: This layer is responsible for abstracting physical resources into virtual resources. The common industry term for this layer is 'Hypervisor'. This layer decouples software from hardware, which enables the software to progress independently from hardware.
5. *VIM (virtualised infrastructure manager)*: This is the management system for NFVI. It is responsible for controlling and managing the NFVI compute, network and storage resources within one operator's infrastructure domain. It is also responsible for collecting performance measurements and events.
 6. *NFV orchestrator*: This creates, maintains and tears down the network services of VNF. If there are multiple VNFs, the orchestrator will enable the creation of end-to-end services over them. NFV orchestrator is also responsible for global management of NFVI resources like compute, storage and networking resources among multiple VIMs in the network. The orchestrator performs its functions by not talking directly to VNFs but through VNFM and VIMs. The VNF manager, VIM and NFV orchestrator, together form the NFV management and orchestration (MANO) entity.
 7. *OSS/BSS (operation support system/business support system)*: OSS deals with network management, fault management, configuration management and service management. BSS deals with customer management, product management, order management, etc. In the NFV architecture, the current OSS/BSS of an operator may be integrated with the NFV management and orchestration, using standard interfaces.

Challenges

What NFV can offer service providers in terms of benefits is clear, but there are challenges for operations and operation

support systems (OSS). Unless addressed appropriately, these challenges can impact a successful transformation. Some network experts argue that a new network management model is needed for NFV and its implied orchestration.

- *Infrastructure challenges*: The infrastructure challenges with NFV transformation are connected to the introduction of new types of components in the network, originating from the technical world and based on industry-standard platforms. These components need to provide telco-grade availability and high performance, while meeting the stringent SLAs that are typical in the communications service providers (CSP) world. Additionally, there is a large installed base of legacy telco equipment. Although much of this current infrastructure will eventually be replaced by virtualised entities, either through an NFV transformation initiative or normal life cycle management, a hybrid environment will always exist. Virtualisation of certain telco equipment won't necessarily be feasible or desirable in every deployment situation.
- *Operational challenges*: A significant operational challenge of the NFV transformation is how to maintain the customer and services views that are tied to the underlying infrastructure. This requires integration within existing OSS/BSS environments and end-to-end automation to enable agility and faster service velocity. By decoupling functions from the infrastructure, NFV warrants new procedures for testing, validation, acceptance and troubleshooting. Another important operational challenge is to manage operations costs while deploying NFV. Factors contributing to this challenge include the complexity associated with managing a hybrid environment of virtualised and legacy equipment and that of managing functions implemented by a distributed set of software entities.
- *Service challenges*: The transformation to NFV will eventually create a CSP infrastructure that is programmable in real-time and is highly automated. In order to take full advantage of the investments in the infrastructure, CSPs will have to revamp the way they offer services to their customers. The challenges in the service domain will mainly be in the area of managing a dynamic service offering and its integration to the underlying infrastructure and real-time analytics.

Open standards for NFV

The standards for NFV are under development by ETSI (the European Telecommunications Standards Institute). Though the concept and benefits of NFV are simple enough, implementing NFV gets complicated. In order to realise the full benefit of NFV, some level of cooperation and interaction between various network solutions providers and network operators is needed.